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Atomic-Scale Phase Composition through Multivariate Statistical Analysis of Atom Probe Tomography Data

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Abstract: We demonstrate for the first time that multivariate statistical analysis techniques can be applied to atom probe tomography data to estimate the chemical composition of a sample at the full spatial resolution of the atom probe in three dimensions. Whereas the raw atom probe data provide the specific identity of an atom at a precise location, the multivariate results can be interpreted in terms of the probabilities that an atom representing a particular chemical phase is situated there. When aggregated to the size scale of a single atom (~0.2 nm), atom probe spectral-image datasets are huge and extremely sparse. In fact, the average spectrum will have somewhat less than one total count per spectrum due to imperfect detection efficiency. These conditions, under which the variance in the data is completely dominated by counting noise, test the limits of multivariate analysis, and an extensive discussion of how to extract the chemical information is presented. Efficient numerical approaches to performing principal component analysis (PCA) on these datasets, which may number hundreds of millions of individual spectra, are put forward, and it is shown that PCA can be computed in a few seconds on a typical laptop computer.

Key words: atom probe tomography (APT), multivariate statistical analysis (MVSA), PCA

Introduction

Multivariate statistical analysis (MVSA) has proven to be an effective approach for extracting the chemical information from large spectral-image datasets in three spatial dimensions. Examples include tomographic spectral imaging with energy dispersive X-ray spectroscopy (EDXS) (Kotula et al., 2006), and time-of-flight secondary ion mass spectrometry (TOF-SIMS) (Smentkowski et al., 2007). In these cases, a complete spectrum is acquired at each point, or voxel, in a regular three-dimensional (3D) array yielding an abundance of data that can be cumbersome to analyze. The assumption underlying MVSA is that a multivariate observation—namely, an n-channel spectrum—is a linear combination of a small number of hidden, or latent, variables. The goal of MVSA is to discover the latent variables, which presumably describe the fundamental spectral characteristics of the substances that make up the sample. If there are p such latent variables with $p \ll n$, MVSA accomplishes a large reduction in the size and dimensionality of the dataset. The presence of correlations among measured variables is essential to the operation of MVSA. It is the redundant information provided by correlated variables that is discarded as the dataset is compressed by MVSA. For a mass spectrometric detection scheme, correlated variables might include, for example, multiple isotopes of a given element or several elements that co-vary in a regular way as in an alloy.

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Implicit in the foregoing discussion is the notion that several variables are measured simultaneously at each spatial location, that is, that the observation is truly multivariate. Atom probe tomography (APT) data, on the other hand, would seem to be fundamentally univariate in nature. Since single atoms are detected one at a time, it is not obvious how the interspecies correlations that MVSA exploits are captured in the data at the atomic level. Nonetheless, there has been recent interest in applying multivariate techniques to APT data (Keenan et al., 2010; Parish et al., 2010; Parish & Miller, 2010). The common approach in the cited works was to partition the sampled volume into a large number of volume elements and to combine all of the ions detected in each individual element into a single mass spectrum. Typically, the volume elements were on the order of 1 nm^3 and contained several to a few tens of detected ions. Likewise, masses were binned to account for the fact that the mass measurements have some imprecision. MVSA was then performed on the aggregated data. Both groups employed the same approach (Keenan, 2009; Smentkowski et al., 2009) whereby principal component analysis (PCA) was used to compress the data and extract the chemical information, and the principal components were, optionally, post-processed to improve interpretability.

MVSA of APT data aggregated in this manner proved successful, but at the cost of losing the high spatial resolution characteristic of atom probes. For a nominal lateral resolution of 0.2 nm, binning to 1 nm^3 implies a factor of 125 loss of resolution on a volumetric basis. In this article we will use simple arguments and simulations to suggest

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that MVSA, in fact, can be successfully applied to APT data aggregated to the native spatial resolution of the instrument. Thus, we can gain the advantages of MVSA, namely, discovering relationships among elements, while retaining full spatial resolution. The arguments turn on the insight that while the atom probe only detects single ions, those can be considered random samples drawn from a local neighborhood defined by the spatial resolution of the instrument. When constructed at the atomic scale, the multivariate model has a natural interpretation. Rather than predicting which specific atom is situated at a particular location, the model predicts the probabilities that atoms of any type are located there. In essence, the chemical-phase composition at the atomic scale is resolved. These principles will be further illustrated by analyzing an APT dataset obtained from a two-phase region of a nickel-base superalloy sample using PCA.

It should be noted that analyzing data binned to the size of single atoms poses significant computational challenges. The number of voxels can easily reach tens to hundreds of millions. Despite the high detection efficiency of atom probes (up to ~60%), less than one ion will be observed, on average, as arising from any single voxel. Consequently, the matrices representing APT mass spectral datasets are extremely sparse, a fact that must be exploited to make the MVSA calculations tractable. Further efficiencies can be achieved by recalling that finding correlations among variables is the central function of MVSA. Since the observation of a single ion in isolation provides no information about interatom correlations, we can restrict our attention to those mass spectra that contain two or more total counts. This will typically be a tiny fraction of the total number of spectra (voxels), yet still a large number in absolute terms. With these considerations in mind, we will discuss a computational approach using standard mathematical tools (e.g., MATLAB, http://www.mathworks.com) that enables the MVSA calculations to be performed on computers with modest capabilities. Using these techniques, PCA of the superalloy dataset below was accomplished in a few seconds on a standard laptop computer.

Materials and Methods

The APT dataset employed in this work was obtained from a sample of a nickel-base superalloy. Non-site-specific APT specimens were prepared using the matchstick method and electropolishing in 10% perchloric acid in acetic acid. Analyses were obtained on a Cameca LEAP 3000X HR in voltage-pulsed mode. This instrument has a detection efficiency of 37% of the ions independent of ion type. Atom positioning precision is better than 1 nm in all locations and can be as high as 0.06 nm in the longitudinal direction (Z axis) of the data (Gault et al., 2010). An overview of atom probe tomography can be found in Kelly and Miller (2007).

Figure 1a shows a 3D rendering of the total ion count for this sample, which was obtained by binning the detected ions into 1 nm^3 cubes. The ions included here were limited

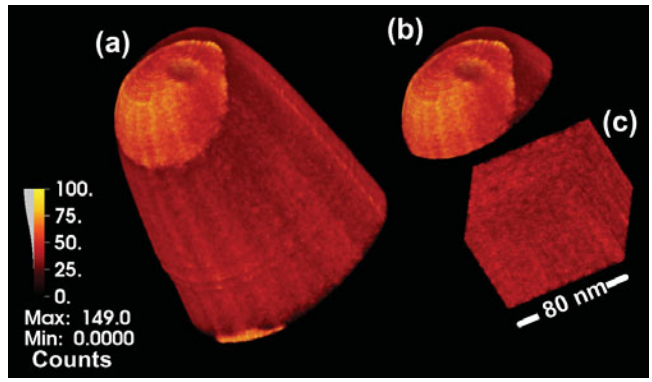

Figure 1. a: Total ion count for the nickel-base superalloy sample when binned to 1 nm in each dimension. **b:** Region of interest is a 39 nm thick two-phase zone. **c:** A volume, representing a single homogeneous phase, was used to characterize the statistical properties of APT data.

to those having a mass-to-charge-state ratio of 100 Da or less. The sample extends 171 nm in the axial dimension, and at its maximum lateral extent, the sample has a diameter of approximately 144 nm. A total of more than 55 million ions were detected. Two regions of interest (ROIs) were selected for further study. A two-phase region shown in Figure 1b was used to test the multivariate approach on a sample having heterogeneous composition. This ROI comprises the top 39 nm of the sample, has a lateral extent of 100 nm at its base, and contains approximately 7.4 million detected ions. For the multivariate analysis, a spectrum at each voxel was constructed by binning all ions at a given location into bins of 0.5 Da width centered on unit and half-unit mass-to-charge-state ratio. The second ROI shown in Figure 1c is a rectangular prism measuring $80 \times 80 \times 56$ nm^3 and was used to evaluate the statistical properties of APT data for a relatively homogeneous material. 12.5 million detected ions were contained in this ROI.

Simple simulations that were performed were based on the physical properties of pure nickel. The relevant properties include the relative abundances of the five naturally occurring isotopes of pure nickel (Rosman & Taylor, 1998), and the face-centered-cubic (fcc) crystal structure of nickel with a lattice parameter of 0.352 nm (Barrett & Massalski, 1980). It was assumed in all simulations that the probability of detecting any given ion was 37%, regardless of the type of ion or its location. Under these conditions, the data are expected to follow the multinomial distribution (Larsen & Marx, 1986) because six possible outcomes (an ion is one of the five isotopes, or no ion is detected), each with a known probability of occurring, span the observation space when looking at all atom positions one at a time. To perform the simulations, a simple multinomial random number generator was developed that was based on random numbers uniformly distributed on the interval from zero to one. Since the probabilities of the six events sum to one, all possible outcomes can be mapped to nonoverlapping subintervals on [0 1], where lengths of the subintervals are equal to the respective probabilities of occurring. An "ob-

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servation" is made by generating a uniformly distributed random number and determining which subinterval includes it.

Multivariate Statistical Analysis

Multivariate statistical analysis techniques generally assume that the chemistry and physics that underlie a spectral dataset can be described by a linear additive model. Given a sample composed of p pure chemical components, elements, or mixtures of elements that vary in the same way with each possessing a characteristic spectrum, the spectrum at any location is the sum of the pure-component spectra weighted by their respective abundances at that location. In the absence of noise,

$$\mathbf{D}_0 = \mathbf{A}_0\mathbf{S}_0^T = \mathbf{T}_0\mathbf{P}_0^T. \tag{1}$$

Here, $\mathbf{D}_0$ is m-voxel $\times$ n-channel matrix of noise-free spectra, $\mathbf{S}_0$ is an n-channel $\times$ p-component matrix whose columns represent the pure-component spectra, and $\mathbf{A}_0$ is an m-voxel $\times$ p-component matrix of abundances. The second equality in equation (1) reflects the fact that the chemical information can also be described in an alternative basis, namely, as principal components. $\mathbf{T}_0$ is an $m \times p$ matrix whose mutually orthogonal columns are called the scores vectors, and $\mathbf{P}_0$ is an $n \times p$ matrix whose orthonormal columns are termed the loadings. The principal-component representation is the unique basis (within its sign) that is orthogonal and serially maximizes that variance that each component explains. In this article, the symbols $\mathbf{T}$ and $\mathbf{P}$ will always refer specifically to components that satisfy the orthogonality and variance maximization constraints of PCA. Arbitrary abundance and spectral factors will be represented by the symbols $\mathbf{A}$ and $\mathbf{S}$, respectively. The principal-component basis is particularly convenient to work with due to its orthogonality, uniqueness, and the fact that efficient algorithms exist to compute the principal components (Jollife, 2002). A simple MATLAB function to accomplish the transformation from an arbitrary factor model to principal components as shown in equation (1) is given by the fPCA function provided in Keenan (2007).

The nickel-base superalloy ROI shown in Figure 1b was analyzed by principal component analysis. In this case, the as-measured raw spectral data contain the underlying chemical information that is now contaminated with experimental error.

$$\mathbf{D} = \mathbf{D}_0 + \mathbf{E}. \tag{2}$$

Here, $\mathbf{D}$ is the $m \times n$ matrix of raw spectral data, and $\mathbf{E}$ is a matrix of random errors having the same dimensions as $\mathbf{D}$, and that is assumed to have zero mean and to be uncorrelated with the systematic part of the data. The principal components of the noisy data and the underlying noise-free chemical model are related through equations (1) and (2) by

$$\mathbf{D} = \mathbf{T}_0\mathbf{P}_0^T + \mathbf{E} = \mathbf{T}\mathbf{P}^T. \tag{3}$$

It is critically important to realize that the principal components of $\mathbf{D}$, $\mathbf{T}$, and $\mathbf{P}$, do not, in general, equal the principal components of the chemical model, $\mathbf{T}_0$ and $\mathbf{P}_0$, which are the ones we are trying to estimate. There are two special cases, however, in which the equality does hold. If $\mathbf{E}$ is negligibly small, it can simply be ignored. As we will show later, noise is, in fact, the dominant source of variation in APT data binned to the atomic scale, so it cannot be ignored. The second case prevails when a single error covariance matrix $\mathbf{C}$ can be used to describe the error in all of the spectra, and $\mathbf{C}$ is equal (or proportional) to the $n \times n$ identity matrix $\mathbf{I}_n$. In this case, the cross product of the raw data matrix can be written as

$$\mathbf{D}^T\mathbf{D} = \mathbf{P}_0(\mathbf{T}_0^T\mathbf{T}_0)\mathbf{P}_0^T + \mathbf{E}^T\mathbf{E} \tag{4}$$

or

$$\mathbf{D}^T\mathbf{D} = \mathbf{P}_0(\mathbf{\Sigma}_0^2)\mathbf{P}_0^T + m\mathbf{I}_n, \tag{5}$$

where the latter expression uses the fact that under the stated conditions the statistical expectation of $\mathbf{E}^T\mathbf{E}$ is $m\mathbf{C} = m\mathbf{I}_n$, and $\mathbf{\Sigma}_0^2$ is a $p \times p$ diagonal matrix containing the eigenvalues of the noise-free underlying model along the diagonal. In the normal case, the raw data cross-product matrix $\mathbf{D}^T\mathbf{D}$ will have full numerical rank—thus, will have n nonzero eigenvalues. If we consider the matrix $\mathbf{P} = [\mathbf{P}_0\ \tilde{\mathbf{P}}_0]$, where $\tilde{\mathbf{P}}_0$ is an orthonormal basis for the model's spectral null space, it is easy to verify that $\mathbf{P}$ diagonalizes both terms on the right-hand side of equation (5).

$$\begin{aligned}\mathbf{P}^T(\mathbf{D}^T\mathbf{D})\mathbf{P} &= \mathbf{P}^T(\mathbf{P}_0\mathbf{\Sigma}_0^2\mathbf{P}_0^T)\mathbf{P} + m\mathbf{P}^T\mathbf{P} \\ &= \begin{bmatrix}\mathbf{\Sigma}_0^2 & \mathbf{0} \\ \mathbf{0} & \mathbf{0}\end{bmatrix}_{n\times n} + m\mathbf{I}_n,\end{aligned} \tag{6}$$

where $\mathbf{0}$ in this equation represents blocks of zeros with conforming sizes that arise due to the orthogonality of $\mathbf{P}_0$ with respect to its null space. This direct calculation shows that the vectors in $\mathbf{P}$ are indeed eigenvectors of the raw data cross-product matrix with corresponding eigenvalues:

$$\mathbf{\Sigma}^2 = \begin{bmatrix}\mathbf{\Sigma}_0^2 & \mathbf{0} \\ \mathbf{0} & \mathbf{0}\end{bmatrix}_{n\times n} + m\mathbf{I}_n = \begin{bmatrix}\mathbf{\Sigma}_0^2 + m\mathbf{I}_p & \mathbf{0} \\ \mathbf{0} & m\mathbf{I}_{n-p}\end{bmatrix}_{n\times n}. \tag{7}$$

We conclude from this analysis that if we sort the eigenvectors of $\mathbf{D}^T\mathbf{D}$ by the magnitude of the eigenvalues in descending order, the p most significant eigenvectors are the model's loading vectors $\mathbf{P}_0$ that we are seeking. The scores can then be estimated simply as

$$\mathbf{T} = \mathbf{D}\mathbf{P}_0. \tag{8}$$

The 0 subscript on $\mathbf{T}$ has been dropped because we cannot distinguish the contribution that the model makes to the scores from that due to collinear noise: substituting equation (3) for $\mathbf{D}$ gives

$$\mathbf{T} = \mathbf{T}_0 + \mathbf{E}\mathbf{P}_0. \tag{9}$$

That is, the scores are equal to the scores of the underlying model plus the projection of the noise into the space spanned by the spectral loadings. It should also be noted that all of the eigenvalues corresponding to noise are equal to m in a statistical sense and that the eigenvalues of the model add to these. Looking for a discontinuity in a plot of

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the sorted eigenvalues provides a basis for selecting the proper number components to model.

APT data, like data from any technique that relies on particle counting, violate the assumption that the noise covariance matrix is proportional to the identity matrix. It is frequently possible, however, to transform the data into a space where the assumption is met. In the present case, it will be shown that when binning at the atomic scale, APT data are approximately Poisson-distributed. For Poisson data, the error covariance matrix can be estimated *a priori* as a diagonal matrix having the elements of the mean spectrum placed along the diagonal (Keenan & Kotula, 2004). Pre- and post-multiplying equation (4) by the inverse of the matrix square root (Golub & Van Loan, 1996) of $\mathbf{C}$ yields an analogue to equation (5):

$$\mathbf{C}^{-1/2}\mathbf{D}^T\mathbf{D}\mathbf{C}^{-1/2} = (\mathbf{C}^{-1/2}\mathbf{P}_0)(\mathbf{\Sigma}_0^2)(\mathbf{C}^{-1/2}\mathbf{P}_0)^T + m\mathbf{I}_n. \tag{10}$$

Eigenanalysis of the weighted data cross-product matrix yields

$$\mathbf{C}^{-1/2}\mathbf{D}^T\mathbf{D}\mathbf{C}^{-1/2} = \underset{\sim}{\mathbf{P}}\mathbf{\Sigma}^2\underset{\sim}{\mathbf{P}}^T, \tag{11}$$

where $\underset{\sim}{\mathbf{P}}$ and $\mathbf{\Sigma}^2$ contain the eigenvectors and eigenvalues, respectively, sorted in order of decreasing significance as indicated by the magnitude of the eigenvalues. Comparing equations (10) and (11), and using the subscript p to indicate the p most significant eigenvalues and eigenvectors, shows that the eigenvectors $\underset{\sim}{\mathbf{P}}_p$ of the scaled data cross product are related through row and column scaling to the loading vectors $\mathbf{P}_0$ corresponding to the physical model. Components in physical space can be computed by inverse scaling as

$$\mathbf{S} = \mathbf{C}^{1/2}\underset{\sim}{\mathbf{P}}_p$$

and

$$\mathbf{A} = \mathbf{D}\mathbf{C}^{-1}\mathbf{S}. \tag{12}$$

The components are written in the generic $\mathbf{A}$ and $\mathbf{S}$ notation because they do not satisfy the constraints of principal components; the scaling in equation (12) destroys the orthogonality of the eigenvectors. These components, however, do span exactly the same space as the principal components and can be easily transformed into principal components by a simple (oblique) rotation with an invertible matrix $\mathbf{R}$. Given the equality of the two factor models, that is, $\mathbf{T}\mathbf{P}_0^T = \mathbf{A}\mathbf{S}^T$, and the orthonormality of $\mathbf{P}_0$, the scores in physical space can be computed as

$$\mathbf{T} = \mathbf{A}(\mathbf{S}^T\mathbf{P}_0) = \mathbf{A}\mathbf{R}. \tag{13}$$

Considering the relationships among spectral bases as expressed in equations (10), (11), and (12), we note that after inverse scaling

$$\mathbf{P}_0\mathbf{\Sigma}_0^2\mathbf{P}_0^T + m\mathbf{I}_n = \mathbf{S}\mathbf{\Sigma}_p^2\mathbf{S}^T. \tag{14}$$

The eigen-decomposition represented by equation (14) shows that the physical-space loadings of the model, the vectors in $\mathbf{P}_0$, are the eigenvectors of the $n \times n$ matrix $\mathbf{S}\mathbf{\Sigma}_p^2\mathbf{S}^T$ with significant-component eigenvalues related by

$$\mathbf{\Sigma}_S^2 \approx \mathbf{\Sigma}_0^2 + m\mathbf{I}_p. \tag{15}$$

Equality holds only approximately because we do not know the true noise covariance matrix but rather are estimating it from the data itself. Recognizing that $\mathbf{S}\mathbf{\Sigma}_p^2\mathbf{S}^T$, which has rank p, is highly rank-deficient allows $\mathbf{P}_0$ to be more efficiently computed as the left singular vectors of the $n \times p$ matrix $\mathbf{S}\mathbf{\Sigma}_p$, which are obtained by singular value decomposition (SVD). Finally, the latter SVD can be computed very efficiently using a kernel approach (Wu et al., 1997) yielding the final result:

$$\mathbf{R} = \mathbf{S}^T\mathbf{P}_0 = (\mathbf{S}^T\mathbf{S})\mathbf{\Sigma}_p\mathbf{U}\mathbf{\Sigma}_S^{-1}, \tag{16}$$

where $\mathbf{U}$ are the eigenvectors and $\mathbf{\Sigma}_S^2$ are the eigenvalues of the small $p \times p$ matrix $\mathbf{\Sigma}_p\mathbf{S}^T\mathbf{S}\mathbf{\Sigma}_p$. We have discussed this rotation at some length because it can be very advantageous from a computational standpoint. Combining equations (12) and (13),

$$\mathbf{T} = \mathbf{D}\mathbf{C}^{-1}\mathbf{S}\mathbf{R}. \tag{17}$$

That is, the principal-component scores can be defined implicitly in terms of the raw data and its spectral mean, the physical-space spectral components and the rotation. Thus, the calculation of the spatial components can be deferred until they are required; unlike the aforementioned fPCA algorithm, no specific spatial information is needed to determine the rotation. One application of this idea is the calculation of scores for a very large spectral image where the complete scores matrix cannot be contained within core memory. Taking the rotation approach greatly facilitates the implementation of an out-of-core algorithm for this calculation. For completeness, we note that the principal-component loadings can be expressed in terms of the inverse rotation:

$$\mathbf{P}_0 = \mathbf{S}\mathbf{R}^{-T}. \tag{18}$$

It should be mentioned that in Keenan and Kotula (2004) scaling was also applied in the spatial domain. This is important in the case of EDXS and TOF-SIMS, where different materials in a sample frequently have vastly different spectral intensities. This is not expected to be the case in APT because the detection efficiency is independent of the type of atom, and atom densities typically do not differ widely with material. Given the poor signal-to-noise of binned APT data, including a huge number of parameters (one per voxel) to model noise in the spatial domain would seem unwarranted and, in our experience, has deleterious effects on the derived components.

The scaling method outlined in the previous paragraph was used to compute PCA for the nickel-base superalloy sample. All calculations were performed with MATLAB, Release 2010a, using functions developed by the corresponding author. The raw mass spectral data are presented to the user after reconstruction in the form of a binary file as a list of position-tagged masses (*.pos file). This is fundamentally a sparse data storage format, and the data were converted

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directly to a matrix in MATLAB sparse format and stored. In cases where subsets of the full data matrix were analyzed, they were extracted from the complete data matrix using standard MATLAB indexing techniques. For instance, the command

```
D=D(sum(D,2)>=2,:)
```

would be used to select only those spectra containing two or more counts from a data matrix **D** organized as m-voxels $\times$ n-mass channels.

All linear algebra calculations involving the data matrix employed sparse-matrix operations. For a review of sparse-matrix techniques with an emphasis on the MATLAB environment, see Davis (2006). One operation, forming cross products involving the data matrix, was implemented as a C++ mex-file to prevent MATLAB from making unnecessary copies of large matrices. The computational times reported here were measured on a Dell Latitude D630 laptop computer with 2 GB of memory running 32-bit Windows XP.

Results and Discussion

Noise Characteristics of APT Data

The nickel-base superalloy ROI shown in Figure 1c provides an opportunity to investigate the noise characteristics of spatially binned APT data. Since this volume contains an essentially homogeneous material, it can be modeled in a simple intuitive way without resorting to multivariate techniques. Specifically, all voxels are assumed to have the same underlying composition whose mass spectrum is given by the mean spectrum over the entire dataset. In terms of the linear model of equation (1), the noise-free data can be modeled as

$$\mathbf{D}_0 = \mathbf{1}_m \boldsymbol{\mu}^T, \tag{19}$$

where $\mathbf{1}_m$ is a vector of ones having m elements (the number of voxels), and $\boldsymbol{\mu}^T$ is the mean spectrum. The amount of variance that the model contributes to the noisy data is equal to the sum of the squared elements of the model data, which can simply be computed as the trace (sum of the diagonal elements) of $\mathbf{D}_0^T\mathbf{D}_0$. Similarly, the total variance in the noisy data can be computed as the trace of $\mathbf{D}^T\mathbf{D}$. Taking the ratio of these two quantities provides an estimate of the fraction of the total variance in the data that is accounted for by the model. For the super alloy ROI binned to 0.2 nm cubes, this ratio is 0.063; 93.7% of the total variance is due to noise alone. Using the approximate equation (15) to estimate $\boldsymbol{\Sigma}_0^2$ suggests that about 6.8% of the variance is due to the model, close to the value 6.3% derived, above, directly from the data. Clearly, MVSA methods, such as PCA, which seek to maximize the amount of variance that the most significant components explain, will find it more profitable to uncover structure in the noise than to discover the underlying chemistry of the sample. As noted in the previous section, the key to achieving good results with MVSA is to remove the structure in the noise prior to performing the multivariate analysis.

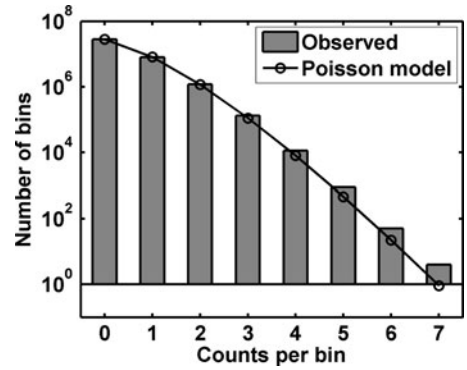

Figure 2. Histogram of the total count distribution for the nickel-base superalloy ROI shown in Figure 1c when binned to 0.2 nm in each dimension. A predicted distribution based on the Poisson model with the same mean number of counts per bin (≈0.29) is also shown.

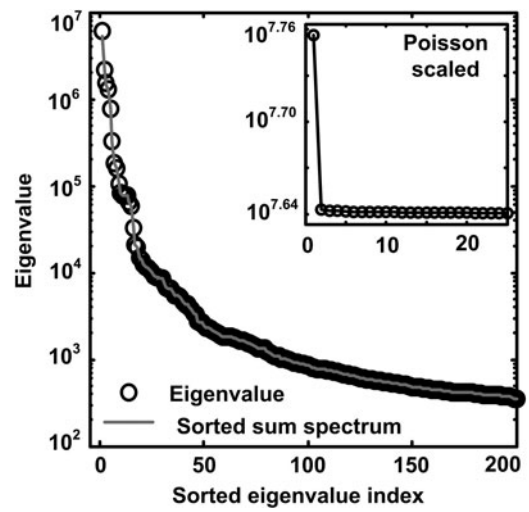

Figure 3. Eigenvalue plot for raw spectral data corresponding to the nickel-base superalloy volume shown in Figure 1c together with a plot of the mean spectrum sorted in descending order (see text for additional detail). The data were spatially binned to 0.2 nm cubes prior to eigenanalysis. The inset shows the first 25 eigenvalues obtained from Poisson-scaled data and clearly suggests the presence of a single significant component.

For the case of Poisson-distributed data, one approach that is effective for eliminating structure in the noise (i.e., making the noise covariance matrix look like the identity matrix) is scaling the data by the inverse square root of the mean spectrum as outlined earlier. Of course, this approach will only be effective to the extent that the data are truly Poisson. Figure 2 shows a histogram of the total counts per voxel for the superalloy ROI of Figure 1c. It is important to note that at this level of binning, only 0.29 counts per spectrum are observed, on average. The distribution of total counts predicted for a Poisson variable having a mean of 0.29 is also displayed in Figure 2 for comparison. The agreement is very good. Figure 3 compares the eigenvalues

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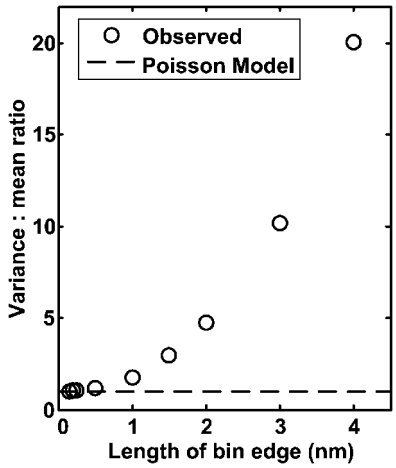

Figure 4. Ratio of the variance to the mean for the total count distribution of the nickel-base superalloy ROI shown in Figure 1c when spatially binned to different extents. For Poisson-distributed data, the variance should equal the mean at all size scales.

obtained from the ROI raw spectral data with the sum of all of the spectra contained in that volume. As noted in Keenan (2007), the eigenvalues corresponding to Poisson noise should equal the summed spectrum. The fact that the sorted summed spectrum agrees almost quantitatively with the eigenvalues indicates, again, that PCA performed on the raw data describes mostly noise. The inset in Figure 3 shows part of the eigenvalue plot after the data have been scaled to account for Poisson noise. The plot clearly suggests the presence of a single significant component, as expected, and the near equality of the eigenvalues corresponding to noise indicates that the method has done a good job eliminating structure from the noise.

One final point that should be mentioned is that APT data follow the Poisson distribution only in the limit that detecting any given atom in a single voxel is a rare event. Figure 4 shows how the ratio of the variance in the total number of atoms per voxel changes with the mean as APT data are spatially binned more coarsely, that is, into bins of larger size. For Poisson-distributed data, the variance should equal the mean independently of the size of the voxel. For APT data, however, the variance is a quadratic function of the mean. Once the bins have reached a couple of nanometers on a side, the data deviate significantly from Poisson. Interestingly, the actual behavior of APT data is exactly opposite to the behavior supposed in Parish and Miller (2010). The origin of the quadratic behavior is understood, and a fuller discussion of APT noise together with a method for dealing sensibly with it during MVSA will be the subject of a separate article.

Simulations

Simple simulations were carried out to investigate the feasibility of using MVSA techniques to extract useful information with atomic size scale resolution from APT data. These calculations simulated the analysis of a pure nickel sample using APT, and parameters included the structure and isotopic composition of pure nickel together with basic performance characteristics of atom probe instruments such as detection efficiency and spatial resolution. In the simplest case, we considered sampling one million atom locations in the nickel structure one site at a time. Given the detection efficiency of the atom probe, taken to be 37% (LEAP 3000X HR, www.cameca.com), there is a 63% probability that no atom of any type will be found at a given location. The remaining possibilities are that we detect a single atom and that its identity is one of the five naturally occurring isotopes of nickel. A five-channel spectrum can be constructed at each location by assigning one channel to each of the possible isotopes and placing a 1 in the channel corresponding to the identity of the observed ion, and zeros in the remaining channels. Note that 63% of the spectra will contain nothing but zeros when we are looking at atom sites one at a time. In the language of MVSA, the five-channel spectrum is a multivariate observation, and the single latent variable we are trying to identify is simply nickel. The variable loadings for the nickel component, in this case, are just the relative abundances of the respective isotopes.

Multivariate data having these characteristics—namely, that each variable in the measurement is limited to taking on a discrete set of possible values, each having a fixed probability—are said to be multinomially distributed (Larsen & Marx, 1986). The multinomial distribution is closely related to the binomial distribution, and, in fact, each individual variable examined separately is binomially distributed. The binomial distribution relates to measurements that can only have two possible outcomes; for instance, an individual atom of Ni is either ^{58}Ni or it is not. For our purposes, the most important property of the binomial distribution is that in the limit of large numbers of observations of low probability events, the binomial distribution is closely approximated by the Poisson distribution (Larsen & Marx, 1986). Thus, while a multinomial random number generator was used to construct the simulations, the data were analyzed assuming they are Poisson-distributed.

Table 1 lists the elements of the most significant principal-component loading vector for the simulation when the atoms are sampled one at a time. For both standard PCA and Poisson-scaled PCA, only the loading corresponding to ^{58}Ni is nonzero. This was to be expected. Since two or more atoms are never observed simultaneously, the data appear uncorrelated ($\mathbf{D}^T\mathbf{D}$ is a diagonal matrix), and since the first principal component tries to account for the most variance in the data, it identifies the most prevalent isotope of nickel, ^{58}Ni. The simple column scaling employed by the Poisson-scaled PCA method cannot induce any correlations among the isotopes, thus yields the same answer.

The situation becomes more interesting when the atoms are sampled two at a time. As shown in Table 1, correlations among all of the isotopes are observed in the first principal-component loading. The Poisson-scaled PCA loading vector quantitatively describes the known relationships among the isotope abundances. Thus, while at most two different isotopes are observed in any single spectrum,

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Table 1. Summary of Results for Simulations Based on the Composition of Pure Nickel.*

	Nominal Mass				
	58	60	61	62	64
Natural abundance	0.6808	0.2622	0.0114	0.0363	0.0093
Simulation abundance	0.6807	0.2620	0.0114	0.0366	0.0093
Sample one atom at a time					
PCA loading 1	1	0	0	0	0
Poisson-scaled PCA loading 1	1	0	0	0	0
Sample two atoms at a time					
PCA loading 1	0.8821	0.1019	0.0031	0.0103	0.0025
Poisson-scaled PCA loading 1	0.6796	0.2628	0.0113	0.0370	0.0093
Lattice binned to 0.2 nm					
PCA loading 1	0.9082	0.0797	0.0024	0.0078	0.0019
Poisson-scaled PCA loading 1	0.6809	0.2620	0.0115	0.0364	0.0092

*The most significant principal-component loading is shown as obtained by PCA both with and without scaling for Poisson noise. The loading is normalized to sum to one to facilitate comparison with the known distribution of the isotopes.

observing many samples of all possible isotope combinations taken pair-wise allows the complete correlation structure to be inferred. It should be noted that standard PCA does a very poor job of describing the relative isotope abundances. For Poisson-distributed data, the estimated variance in a measurement is equal to its mean. From the standpoint of simply accounting for variance, it is more advantageous to account for large-magnitude noise in a major spectral feature than variance due to a small spectral feature that has chemical significance. Consequently, the first principal-component loading weights ^{58}Ni much more heavily than its natural abundance would warrant.

The foregoing discussion suggests that MVSA is feasible when APT data are spatially binned to the size of two atoms. In fact, the final simulation demonstrates that MVSA can be applied successfully when APT data are binned to the size of a single atom or smaller. The basic idea is that while two atoms obviously cannot occupy the same physical location, it can *appear* that more than one atom resides at the same place when the data are binned. There are two reasons for this. First, based on the fcc crystal structure of nickel and its lattice parameter, 0.352 nm, it is simple to estimate (Barrett & Massalski, 1980) that the diameter of a nickel atom is approximately 0.25 nm. However, a cube whose sides are only 0.176 nm long (half the lattice parameter) and parallel to the sides of the unit cell, and centered on a tetrahedral hole, will contain the centers of the four atoms forming the tetrahedral hole at vertices of the cube. Thus, any cube of this dimension or greater randomly placed on the lattice will have nonzero probabilities of containing any number up to four atoms. Second, the imperfect precision with which the atom probe locates an atom can make it appear that more than one atom resides at a given position.

The final simulation assumes that atoms in the nickel lattice are located with a Gaussian root-mean-square precision of 0.2 nm in the lateral dimensions and 0.1 nm in depth (Gault et al., 2010). For this simulation, the atoms were binned into cubes that are 0.2 nm on a side. The simulated lattice was 76 × 76 × 52 nm in size (37,544,000 voxels) and a total of 10,193,268 atoms were counted. Thus, the probability of finding an atom of any type in any given voxel was only 27%. Figure 5 displays a histogram of the total number of atoms detected per bin. As expected, the vast majority of the bins contain either zero or one count. However, as many as six atoms are found in a single voxel, and over 1 million voxels contain two or more atoms. While this is only a few percent of the total number of voxels, it is a large number in absolute terms and allows good statistical estimates to be made. The histogram is compared with the distribution expected from Poisson-distributed data, and the agreement is very good as shown in Figure 5. We also note the close similarity of this histogram with the corresponding plot made for the superalloy cube ROI and shown in Figure 2. The first principal component of the lattice simulation obtained by MVSA is included in Table 1, and again, the Poisson-scaled PCA yields quantitatively correct estimates of the nickel isotope distribution.

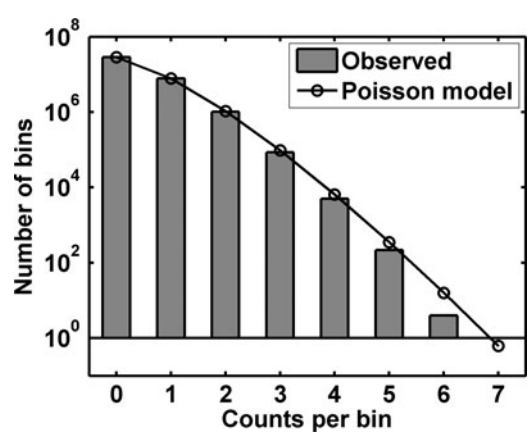

Figure 5. Histogram of the total count distribution for the nickel-lattice simulation with the data binned to 0.2 nm in each dimension. A predicted distribution based on the Poisson model with the same mean number of counts per bin (≈0.27) is also shown.

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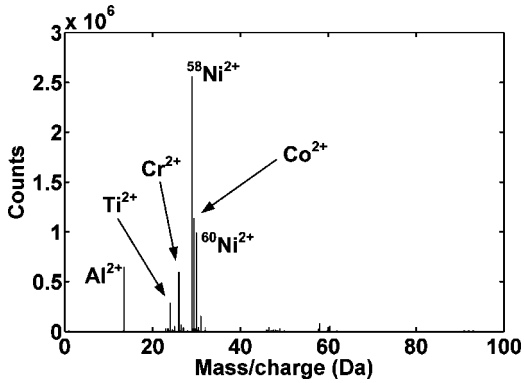

Figure 6. Total ion spectrum for the nickel-base superalloy ROI shown in Figure 1b. The mass spectra are binned to a resolution of 1/6 Da.

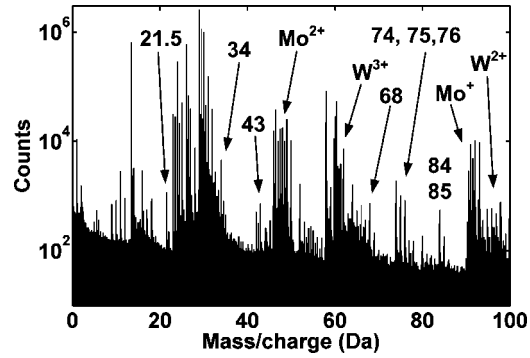

Figure 7. Total ion spectrum from Figure 6 for the nickel-base superalloy replotted on a logarithmic scale.

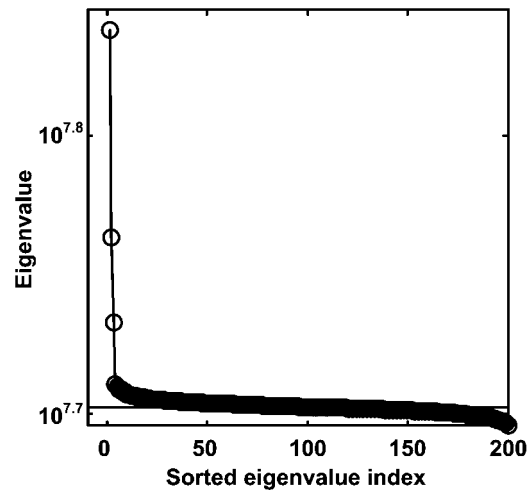

Figure 8. Eigenvalue plot for the nickel-base superalloy two-phase region suggesting the presence of three components. The horizontal line is located at 50,381,604, which is the number of voxels in the dataset and the predicted asymptotic "noise floor" [see equation (7)].

It is interesting to consider exactly how the MVSA model should be interpreted when modeling the composition of a material at the atomic size scale. The native atom probe model of a material predicts that a specific type of atom is present at a particular spatial location. The multivariate model, on the other hand, only predicts that an atom characteristic of a specific phase will be present there. In the context of the nickel-lattice simulation, atom probe analysis may find that a ^{58}Ni atom resides at some particular site; the MSVA model will only predict that some isotope of Ni will be present there, and that the likelihood of any specific isotope residing at that location is given by its relative natural abundance.

MVSA of the Nickel-Base Superalloy Two-Phase Region

Based on the total ion count for the nickel-base superalloy two-phase ROI shown in Figure 1b, it is evident that there are at least two different types of voxels having slightly different total ion intensities. Thus, this sample should provide a test of MVSA's ability to discover multiple components in sparse APT data. The mean spectrum of the entire ROI is shown in Figure 6. The mass spectrum appears relatively simple, consisting largely of elements known to be present from the bulk composition. A more revealing look at the mean spectrum is obtained by replotting it on a logarithmic scale as shown in Figure 7. Clearly, a wealth of information is present in APT data; not immediately clear is how to interpret it.

Eigenanalysis was performed on data spatially binned to 0.2 nm cubes and scaled to account for Poisson statistics. The eigenvalue plot in Figure 8 suggests the presence of three significant components. Referring back to equation (7), eigenvalues corresponding to noise should be equal and have a numerical value equal to the number of voxels if we have properly dealt with the noise. A line corresponding to the number of voxels is shown on the eigenvalue plot and, on average, the noise eigenvalues correspond to the number of voxels. Note that we do not expect exact equality. Since we do not know the true noise covariance matrix, we must estimate it from the data. The successive eigenvalues are then weighted by the degrees of freedom associated with their respective estimates (Malinowski, 1987), which leads to the slight slope the sorted eigenvalues exhibit in Figure 8.

The three most significant principal-component loadings obtained with the Poisson-scaled PCA method are displayed in Figure 9. The first two loadings describe the compositional similarities and differences among the major constituents of the two-phase region. The third loading describes an unexpected set of species that, as will be shown below, are found at the interface. This component is describing two different things. First, there is an enhanced abundance of singly charged metal ions relative to the doubly charged ions predominant in PC1 and PC2 loadings, which are associated with the bulk phases. Second, numerous metal oxides are found that are not present in PC1 and PC2 loadings. Taken together, these three PCs explain most of the unidentified mass peaks in the mean spectrum.

While the principal-component loadings describe the spectral characteristics of the various chemical phases, the corresponding scores give information about their respec-

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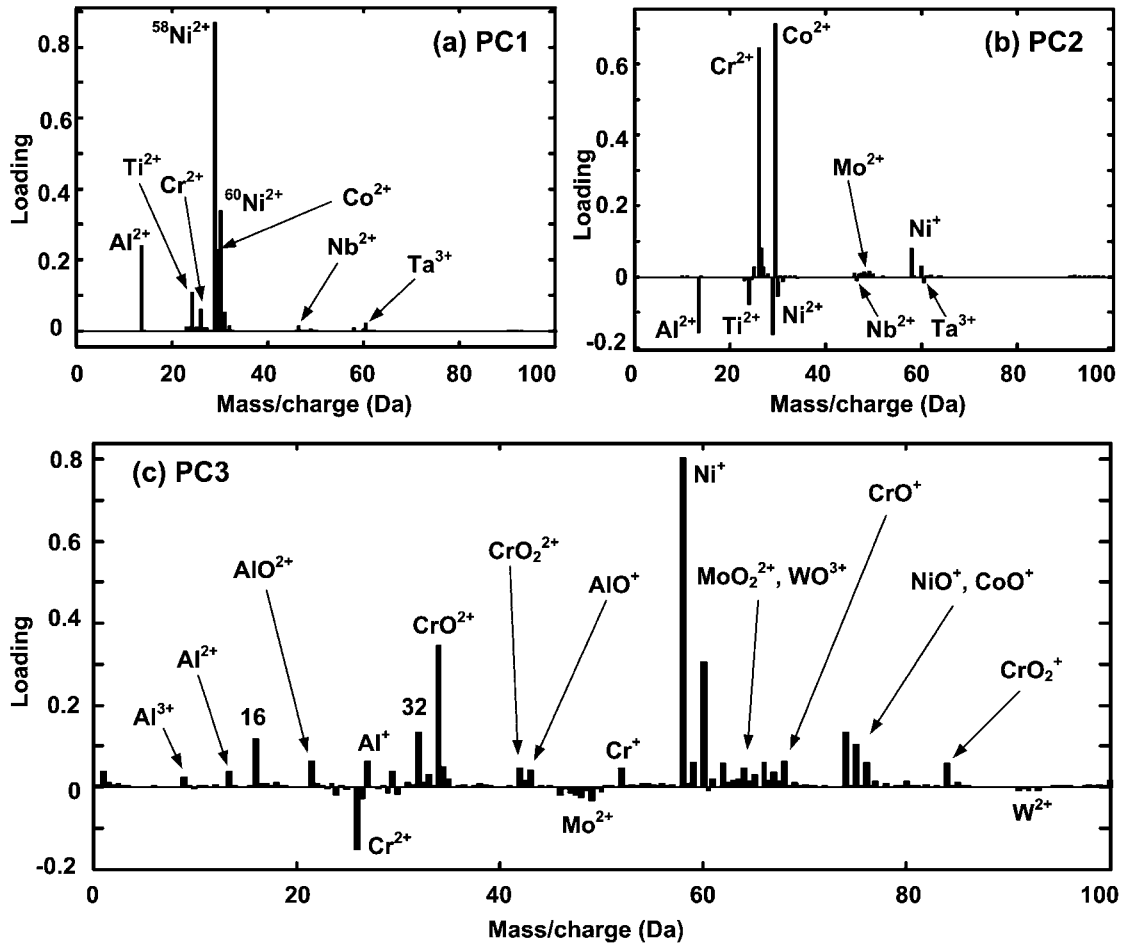

Figure 9. The three principal-component loadings obtained by the Poisson-scaled PCA method for the nickel-base superalloy two-phase region.

tive locations. Figure 10a shows the superalloy two-phase region of Figure 1b after reorienting the view to be more closely aligned with the axis of the sample. Figure 10b presents the three principal-component scores as a color overlay with the same orientation. The resemblance between the two images is striking, and the three score components clearly correlate with the total ion intensity observed in the raw data.

In the present case, it is likely that standard image processing techniques could have been used to segment the total ion image (Fig. 10a) into three (or more) distinct groups of voxels, and each group of voxels could have been studied individually. Of course, performing such processing implies some foreknowledge about the sample that permits an intelligent choice of segmentation criteria—specifically, what masses are important to differentiating chemical phases in the sample. This type of approach is prone to introducing bias into the results, and to missing small but potentially important spectral features of which the analyst may not be aware. Much of the power of MVSA resides in its ability to use all of the collected information to very rapidly discover the number of components, their spectral characteristics, and spatial distributions in an automated way without preconceived notions about what should be present and where. Establishing the fact that all of the oxide species vary together at the interface, for instance, would be tedious at best using univariate (manual) data analysis methods given the very low levels of these species present in the sample. MVSA can also find subtle variations in chemistry that would be difficult to characterize otherwise (Smentkowski et al., 2009). A comparison of the loadings in Figures 9a and 9b shows that there are very small quantities of niobium and tantalum present in the superalloy, and that they are more prevalent in the Ni-rich phase than the Cr/Co-rich phase. This may or may not be important to the performance of the material; what is important is that this information is made available to the analyst for further evaluation. For a more complete discussion of automated analyses, the power of MVSA and other topics addressed in this paragraph, see Kotula et al. (2003).

This article has only been concerned with extracting the chemical information at the atomic size scale from large APT datasets in a maximally parsimonious way. PCA was our choice of multivariate method because it is guaranteed that no p-component factor model can beat PCA in a least-squares fitting sense, and because efficient algorithms for computing PCA are commonly available. The drawback of PCA is that because the constraints imposed on the

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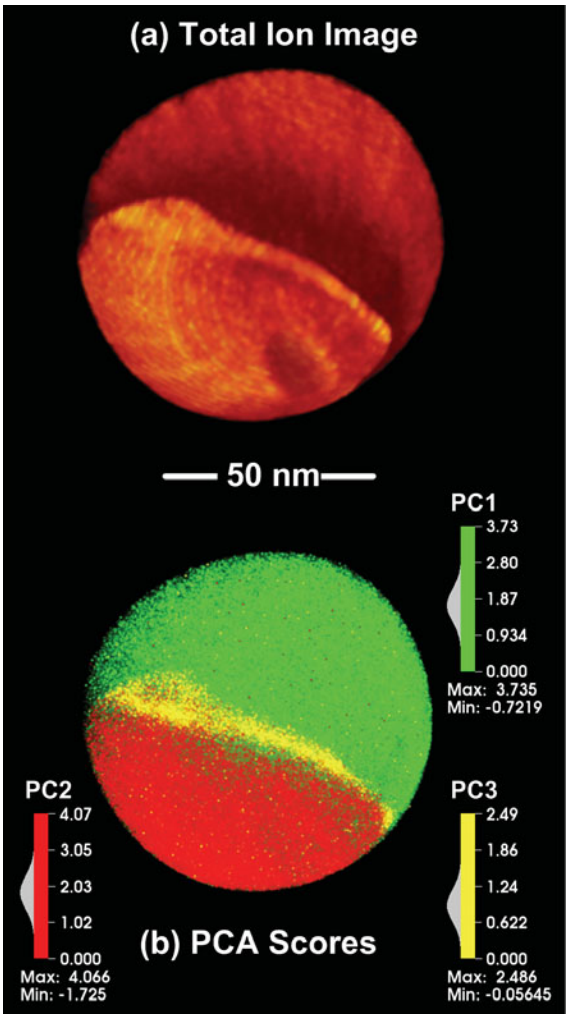

Figure 10. The three principal-component scores obtained by the Poisson-scaled PCA method for the nickel-base superalloy two-phase region displayed as a color overlay. **a:** Total ion image of the ROI from Figure 1b reoriented to look down the axis of the sample. **b:** The PCA scores in the same orientation.

principal components, the orthogonality and variance maximization properties, are not based on physical principles, the scores and loadings derived by PCA frequently appear abstract making them difficult to interpret. The negative spectral features in the loading vectors shown in Figure 9 are examples of this type of abstract or unphysical character. Several types of post-processing steps can be employed in an attempt to make the components more easily interpretable in physical terms. Multivariate curve resolution (MCR; see the MCR homepage, http://www.ub.edu/mcr/web_mcr/welcome.htm, by Romà Tauler, Anna de Juan, and Joaquim Jaumot for a wealth of information about MCR, links to the literature and software) has been employed successfully with large 3D spectral images (Kotula et al., 2006; Smentkowski et al., 2007) to compute components that are nonnegative. Abstract factor rotations have also been employed to transform the principal components into a more physically realistic representation (Keenan, 2009; Smentkowski et al., 2009).

The latter approach was the one taken by Parish and Miller (2010) in their article analyzing APT data, with a method they referred to as "spatial simplicity principal component analysis (SS-PCA)." Unfortunately, the mathematics of the method were incorrectly described in that article. Based on purely mathematical grounds, the nonorthogonal spectral components they show could not have been obtained by rotating a matrix of orthonormal loading vectors with an orthogonal matrix derived from a rotation of the principal-component scores as they suggest. The argument turns on a subtle but crucially important difference between the rotational properties of orthogonal scores and orthonormal loadings. The interested reader is referred to the original literature (Keenan, 2009) for further elucidation.

Computational Considerations

Atom probe datasets can be truly huge when spatially binned to the size scale of a single atom. The dataset corresponding to the complete nickel-base superalloy sample in Figure 1a contains over 56 million detected ions. As noted earlier, the probability of detecting any type of ion in any given voxel with 0.2 nm sides is about 0.3; thus, we can infer that there are almost 190 million voxels within the actual volume of the sample, ignoring the void volume in the circumscribing box. A double-precision, dense-matrix representation of this dataset with a 200-channel mass spectrum at each voxel would require about 280 Gbytes of computer storage. Clearly, performing MVSA would pose a daunting challenge. Fortunately, the raw APT data are presented to the user in a sparse format, and it is possible, in fact essential, to maintain and exploit the sparsity throughout the entire analysis process.

Parish and Miller (2010) describe in detail the procedure for mapping position-tagged ion masses contained in the APT *.pos-file to indices in a four-dimensional (4D) array (three spatial and one spectral). The indices represent the locations of the spatial and spectral bins having user-defined sizes. They further describe how the 4D array is unfolded into a matrix for linear algebra operations. We will not repeat that discussion here. Parish and Miller then create a dense $m \times n$ matrix and accumulate the observed ion detections into the appropriate bins. An enormous savings in storage can be accomplished by realizing that the two d-vectors that contain the matrix row and column indices, respectively, for each of the d detected ions, together with a d-vector of ones reflecting the fact that each set of indices tags a single ion, is simply the standard representation of a sparse matrix in triplet format (Davis, 2006). In Matlab, the "sparse" command sums duplicate entries in the triplet-formatted list and converts the matrix into so-called compressed-column format, which is suitable for performing efficient sparse linear algebra operations. The storage requirements for the compressed-column format are $2 \times d$ plus the number of columns in the matrix. For reasons related to how Matlab performs linear algebra, we actually store the data matrix $\mathbf{D}$ as its transpose; thus, storage is

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Table 2. Dataset Sizes and Times to Compute PCA for Four Different Subsets of the Nickel-Base Superalloy Dataset.*

	Subset			
	a	b	c	d
Number of voxels	50,381,604	50,381,604	448,907,960	448,907,960
m	6,254,136	1,007,266	7,178,898	7,178,898
n	200	200	200	1,200
Nonzero elements	7,125,018	1,878,148	13,186,663	13,803,309
Number of atoms	7,404,993	2,158,123	15,226,812	15,233,703
Average read time (s)	2.65	0.70	6.59	7.56
PCA time: all PCs (s)	0.79	0.18	0.97	8.37
PCA time: 64 PCs (s)	0.87	0.27	1.01	3.62

*The value of m indicates the number of spectra in the subset. (a) ROI of Figure 1b containing all of the nonzero spectra, (b) same dataset but containing only the spectra having two or more total counts, (c) complete sample and spectra containing two or more counts, and (d) complete dataset with six times higher spectral resolution.

required for $2 \times d + m$ elements. This compares with $m \times n$ elements for the dense-matrix representation. For the superalloy sample including the enclosing box, this represents a factor of 160 less storage.

Additional savings can be realized when data are binned to the atomic size scale because most of the spectra contain zero counts and can be ignored as they provide no useful information. However, indices indicating the locations of the nonzero spectra must be retained to enable an accurate rendering of the 3D scores. Perhaps not as obvious is the fact that all spectra containing only a single count can also be safely ignored because a single count yields no information about the interchannel correlations that are central to MVSA. Table 2 lists several size metrics for a few different subsets of the nickel-base superalloy dataset. Table 2 (a) and (b) refer to datasets corresponding to the ROI in Figure 1b. Subset (a) includes all spectra containing at least one count, and subset (b) is that subset of spectra having two or more counts. Subsets (c) and (d) refer to the whole sample shown in Figure 1a and include only those spectra containing two or more total counts. They differ in spectral resolution. Spectra are binned to a resolution of 1/2 Da in subset (c); subset (d) has a spectral resolution of 1/12 Da. The memory required to hold the data matrix describing the largest subset, subset (d), is a manageable 265 Mbyte. The time required to convert the raw *.pos-file to a Matlab-format file for subset (d) was about 2 min on a Dell Precision 690 Workstation with 16 Gbytes of memory and running 64-bit Windows XP. The conversion process has a large memory overhead due largely to Matlab limitations and would be better accomplished at the instrument. After conversion, all MVSA calculations were completed on a standard laptop computer with 2 Gbytes of memory and running 32-bit Windows XP.

PCA was computed for each of the subsets in Table 2. In each case, the eigenvalue plot was similar in appearance to Figure 8 and suggested the presence of three significant components. The principal-component loadings for the component describing the interface region are compared in Figure 11. The loadings shown for subsets (a) and (b) are visually indistinguishable even on a greatly expanded scale. This validates the idea that spectra having only a single count can be ignored from the standpoint of MVSA. The loading in Figure 11c differs in very subtle ways from the previous ones. This is to be expected because this subset contains many more spectra corresponding to the nickel-rich phase. Interestingly, "diluting" the interface pixels in this way has no impact on the quality of the multivariate results for this component. The times required to compute PCA for these three subsets are also included in Table 2. In each case, the actual computation time was one second or less, and less than the time needed to read the data into the computer memory from disk. For all intents and purposes, computing the principal components of the atom probe data at the atomic size scale is a no-cost operation.

The final subset shown in Table 2 (d) and Figure 11d considers the effect of increasing the spectral resolution, which may be useful to the analyst when making mass assignments. From a sparse matrix storage standpoint, the width of a mass bin is irrelevant when only one or a few ions are detected in each spectrum. Thus, we should be able to bin the spectra to the native mass precision of the atom probe with little effect on the memory requirements. That this is indeed the case can be seen comparing columns (c) and (d) in Table 2. The predominant cost of increased spectral resolution is reflected in the time required to compute the eigenanalysis of the data cross-product matrix, an $n \times n$ matrix. The time required to compute the complete eigen-decomposition is expected to scale as n^3 (Golub & Van Loan, 1996). This problem can be mitigated by computing only a few of the most significant eigenvalues and eigenvectors. This is the approach taken by the AXSIA spectral image analysis program (Kotula et al., 2003) written by one of the present authors (M.R.K.), where only the 64 most significant eigenvalues are considered. This same approach and number was also used by Parish and Miller (2010). The bottom two lines in Table 2 compare the PCA computation times when either all or only 64 of the eigenvalue/eigenvector pairs were computed. For relatively low spectral resolution, there is no noticeable difference.

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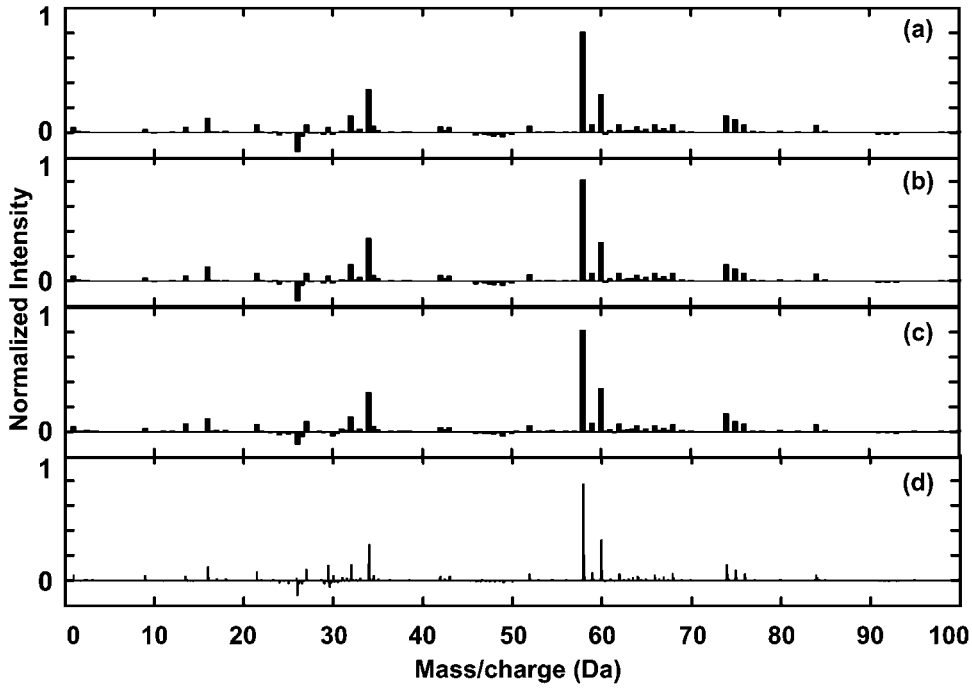

Figure 11. Principal-component loading vectors corresponding to the nickel-base superalloy two-phase interface for the four subsets described in Table 2. Subset (d) has six times higher spectral resolution than subsets (**a**)–(**c**). See Figure 9c for mass assignments.

Sizable time savings can be realized with higher resolution spectra, however. The high-resolution loading shown in Figure 11d, aside from the improved resolution, appears very similar to loadings obtained from the other subsets.

It is important to note that the multivariate analysis results obtained from the ROI differ in no significant way from those derived from the complete dataset. In the latter case, no assumptions were made as to where the interesting voxels were located or which masses provided the most insight into the composition of the sample. All locations and all masses were included in the analysis and treated equivalently. The ability of MVSA to extract the chemical information without the bias that might be introduced by an analyst's choice of ROI or channel selection is one of its greatest strengths. Given that MVSA, in this case, exacts virtually no cost suggests that it should become an important element of the analyst's toolbox.

One final point concerns the explicit computation of the principal-component scores in a way that is usable by 3D visualization tools. In the examples above, we computed the nonzero scores and maintained a list of indices showing where the nonzero scores were located. For display purposes, it may be necessary to supply the full scores matrix, including the locations where the scores are zero, to the software. In this case, the storage requirements for the scores may easily exceed that of the data matrix itself. This is the significance of separating the task of computing the loadings from the task of computing the scores in the discussion leading up to equation (17). Given a set of spectral basis vectors that span the spectral space of a raw data matrix, the rotation matrix that transforms them into principal-component loadings, and an estimate of the noise covariance matrix, scores can be computed directly from the raw data. One might envision, for instance, a process that takes a stream of raw spectra as an input and outputs the corresponding scores. An important point is that the loadings do not need to be derived from the same dataset as the scores, they only need to span the same space. Thus, depending on the needs of the display software, scores at an appropriate spatial resolution can be estimated from loadings obtained at a different resolution without having to recompute the PCA factorization.

Summary

We have demonstrated for the first time the ability to perform MVSA on APT data at the full spatial resolution of the atom probe. Given that raw APT data consist of a list of position-tagged single ion detections, it would at first seem unlikely that the interelement correlations that MVSA exploits can be discovered on the size scale of a single atom. However, while two atoms certainly cannot simultaneously occupy the same physical location, the imperfect precision with which the atom probe locates an atom can make it appear that more than one atom resides at a given position. That is, the single atoms that the atom probe detects can be considered random samples drawn from a local neighborhood defined by the spatial resolution the instrument. Viewed this way, it would seem plausible that multivariate statistical techniques could extract useful information from APT data in an automated and unbiased manner. Specifically, MVSA should be able to estimate the chemical compositions of

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neighborhoods on the atomic size scale without preconceived notions of what types of atoms to look for or where to look for them.

Using simple simulations, the principles outlined above were demonstrated. Additionally, MVSA was successfully applied to a real sample, a two-phase zone in a nickel-base superalloy. Besides the chemical phases expected based on the bulk composition of the superalloy, an unexpected interface that is enriched in oxygenated species was discovered.

Properly accounting for noise is critical to the success of MVSA. Variance-based MVSA methods, such as PCA, attempt to account for variance in the dataset in a maximally parsimonious way, with no regard to whether that variance arises from informative signals or random noise. For APT data minimally binned to the atomic size scale, the signal-to-noise ratio is very low. The most prevalent species in the two-phase sample, ^{58}Ni, is only found in about 5 out of every 100 voxels, on average, when the data are binned into cubes 0.2 nm on a side. Here we show that APT data are approximately Poisson-distributed when binned to this level; thus, the estimated standard deviation for the ^{58}Ni measurement is the square root of the mean, or about 0.22. In other words, the signal-to-noise ratio is significantly less than 1. Without properly accounting for the statistical nature of the noise, MVSA will find only noise. A procedure for dealing with Poisson-distributed data was developed and shown to successfully discriminate chemical information from noise in the case of real APT data aggregated at the atomic scale. Interestingly, APT data binned more coarsely, on the order of 1 nm or more, start to deviate significantly from Poisson. Details of the noise characteristics for coarsely binned APT data and a method to account for this noise sensibly in a multivariate analysis will be the subjects of a separate article.

It seems remarkable that MVSA can be successfully applied to a spectral image when each spectrum contains much less than one count, on average (0.15 total counts per voxel for the two-phase sample). However, the paucity of information in any single spectrum is more than counterbalanced by the vast number of spectra, tens of millions or more, in the dataset. Handling that number of spectra does pose significant computational challenges. Over 50 million 200-channel mass spectra comprise the nickel-base superalloy two-phase ROI dataset. Simply to represent that dataset by a double precision dense matrix would require over 75 Gbytes of memory. We have presented a computational approach that maintains the data in a sparse format from generation through MVSA. Using these techniques, MVSA at the atomic size scale can be accomplished in seconds on computers with modest capabilities. Thus, the feasibility of using MVSA as a routine tool for the analysis of APT data is apparent.

References

Barrett, C.S. & Massalski, T.B. (1980). *Structure of Metals*. Oxford, UK: Pergamon Press.

Davis, T.A. (2006). *Direct Methods for Sparse Linear Systems*. Philadelphia, PA: SIAM.

Gault, B., Moody, M.P., De Geuser, F., La Fontaine, A., Stephenson, L.T., Haley, D. & Ringer, S.P. (2010). Spatial resolution in atom probe tomography. *Microsc Microanal* **16**(1), 99–110.

Golub, G.H. & Van Loan, C.F. (1996). *Matrix Computations*. Baltimore, MD: The Johns Hopkins University Press.

Jollife, I.T. (2002). *Principal Component Analysis*. New York: Springer-Verlag.

Keenan, M.R. (2007). Multivariate analysis of spectral images composed of count data. In *Techniques and Applications of Hyperspectral Image Analysis*, Grahn, H.F. & Geladi, P. (Eds.), pp. 89–126. Chichester, UK: John Wiley & Sons, Ltd.

Keenan, M.R. (2009). Exploiting spatial-domain simplicity in spectral image analysis. *Surf Interface Anal* **41**, 79–87.

Keenan, M.R. & Kotula, P.G. (2004). Accounting for Poisson noise in the multivariate analysis of ToF-SIMS spectrum images. *Surf Interface Anal* **36**(3), 203–212.

Keenan, M.R., Smentkowski, V.S., Ulfig, R.M., Oltman, E., Larson, D.J. & Kelly, T.F. (2010). Multivariate statistical analysis of atom probe tomography data. *Microsc Microanal* **16**(S2), 270–271.

Kelly, T.F. & Miller, M.K. (2007). Atom probe tomography. *Rev Sci Instrum* **78**, 031101.

Kotula, P.G., Keenan, M.R. & Michael, J.R. (2003). Automated analysis of SEM X-ray spectral images: A powerful new microanalysis tool. *Microsc Microanal* **9**(1), 1–17.

Kotula, P.G., Keenan, M.R. & Michael, J.R. (2006). Tomographic spectral imaging with multivariate statistical analysis: Comprehensive 3D microanalysis. *Microsc Microanal* **12**, 36–48.

Larsen, R.J. & Marx, M.L. (1986). *An Introduction to Mathematical Statistics and Its Applications*. Englewood Cliffs, NJ: Prentice-Hall.

Malinowski, E.R. (1987). Theory of the distribution of error eigenvalues resulting from principal component analysis with application to spectroscopic data. *J Chemometrics* **1**, 33–40.

Parish, C.M., Capdevila, C. & Miller, M.K. (2010). Applying multivariate statistical analysis to atom probe tomography. *Microsc Microanal* **16**(S2), 1858–1859.

Parish, C.M. & Miller, M.K. (2010). Multivariate statistical analysis of atom probe tomography data. *Ultramicroscopy* **110**, 1362–1373.

Rosman, K.J.R. & Taylor, P.D.P. (1998). Isotopic compositions of the elements 1997. *Pure Appl Chem* **70**(1), 217–235.

Smentkowski, V.S., Ostrowski, S.G., Braunstein, E., Keenan, M.R., Ohlhausen, J.A. & Kotula, P.G. (2007). Multivariate statistical analysis of three-spatial-dimension TOF-SIMS raw data sets. *Anal Chem* **79**(20), 7719–7726.

Smentkowski, V.S., Ostrowski, S.G. & Keenan, M.R. (2009). A comparison of multivariate statistical analysis protocols for ToF-SIMS spectral images. *Surf Interface Anal* **41**, 88–96.

Wu, W., Massart, D.L. & de Jong, S. (1997). The kernel PCA algorithms for wide data. I. Theory and algorithms. *Chemometrics Intell Lab Syst* **36**(2), 165–172.

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